Capstone Defense — Speaker Notes / Rehearsal Script

Geospatial Flood-Risk Modelling using Climate Data and Hidden Markov Models in Rwanda · Kellen Murerwa

Target: ~10–12 minutes. One slide ≈ 30–60 seconds.

Slide 1. Title slide

Good morning. My name is Kellen Murerwa and my capstone is a geospatial machine-learning and Hidden Markov Model framework that estimates daily flood-pressure for the Nyabugogo-Nyabarongo corridor in Kigali. Over the next ten minutes I will take you through the problem, the real data, the models, and an honest evaluation. (~45s)

Slide 2. Problem & Objective

Kigali floods repeatedly after heavy rain. The official hazard maps are valuable but static: they tell you WHERE flooding is possible, not WHEN or how WIDELY pressure builds after rainfall. My objective is a time-aware decision-support tool that estimates Low, Moderate or High flood-pressure each day and is validated against the official flood polygons. It complements, not replaces, early-warning systems.

Slide 3. Study Area & Grid

I work on a 250 m grid of 729 cells over the Nyabugogo-Nyabarongo corridor, daily from 2018 to 2024 — about 1.86 million cell-day records. I deliberately shifted the study box south so it sits on the real official flood zone. The map on the right is my model's 2024 high-pressure frequency, with the official polygons outlined in blue.

Slide 4. System Architecture (four layers)

The system has four layers: real data; a label layer; the models; and validation plus delivery. Each is a separate reproducible module, all driven by a single main.py entry point, so the whole pipeline reruns end to end.

Slide 5. Real Data Sources

Every predictor is real and retrieved live — nothing is assumed. Rainfall from Open-Meteo ERA5, elevation and slope from SRTM, rivers, roads and buildings from OpenStreetMap, and the official flood polygons from the government portal, geodata.rw. To prove it is real: this is a live ERA5 pull for my area for January 2024 — 131.2 mm of rain, with the actual daily values. The official layer flags 136 of my 729 cells as flood-prone.

Slide 6. Features & Labels (why not circular)

This is the question I most expect from you, so let me address it directly. My label is a flood-pressure score from rainfall and terrain. If I stopped there, the label would be a formula the model could simply reverse-engineer — and indeed an early version scored 0.99 F1, which is a warning sign, not a success. So I added an unobserved drainage-and-noise term the model cannot see, which drops performance to a realistic 0.83. Flood-pressure is a derived risk state, not a confirmed flood event.

Slide 7. Models & Evaluation Protocol

I benchmark against two naive baselines so the machine learning has to earn its place, then interpretable models, then Random Forest and XGBoost with SHAP, and finally a Hidden Markov Model for the time dimension. Crucially I evaluate on a temporal hold-out: I train on 2018 to 2023 and test on the completely unseen year 2024 — so this is generalisation, not memorisation.

Slide 8. Results — 2024 hold-out

On 2024, Random Forest is best with a macro-F1 of 0.831 and high-pressure recall of 0.867 — both above the 0.75 and 0.80 targets. XGBoost is close behind. Both baselines are clearly beaten — the rainfall-only baseline reaches just 0.62 — which proves the geospatial context adds real value. The confusion matrix shows strong Low and High detection; most error is on the transitional Moderate class, which is expected.

Slide 9. Explainability (SHAP)

SHAP confirms the model is hydrologically sensible: 3- and 7-day rainfall dominate, followed by low elevation and closeness to a river. That is exactly the physical story of urban flooding, and it makes the model auditable rather than a black box.

Slide 10. HMM — temporal dynamics

The HMM captures persistence. The self-transition probabilities are about 0.83, 0.80 and 0.91, meaning once a cell enters high pressure after sustained rain it tends to stay there, and it moves through neighbouring states rather than jumping. Next-step accuracy is 0.45 against a random baseline of 0.33.

Slide 11. Spatial Validation vs Official Polygons

Here is my most honest slide. My original target was 70% of predicted-high cells inside the official polygons. But that layer covers only 19% of my corridor, so 70% is mathematically impossible unless I make the label circular. I therefore reframed it to enrichment and an odds ratio. Predicted high-pressure is 1.59 times more concentrated in official zones than chance, and official cells are flagged high 1.84 times more often than others — and the model never saw those polygons in training. It rediscovers the official zone and extends it, which is the core contribution.

Slide 12. Streamlit Inspection Dashboard

Everything is delivered through a Streamlit dashboard. You pick a date, see the predicted map, overlay the official polygons, and click any cell to see its rainfall history and the model's probabilities. It turns a model into a tool a city officer can actually use.

Slide 13. ERA5 vs CHIRPS vs Meteo Rwanda

On data choice: I use ERA5 because it is the only source I can retrieve reproducibly and keylessly, as I just demonstrated live. CHIRPS is scientifically preferable for African rainfall, but I actually tested two access routes — the IRI Data Library and ClimateSERV — and both failed without heavy tooling, so I document it as a validation extension. Meteo Rwanda is the gold standard but has no open API and needs a formal request.

Slide 14. Limitations & Future Work

I am clear about limitations: my labels are flood-pressure, not confirmed events; ERA5 is coarser than a dense gauge network; and the official polygons are low-resolution. Future work is incident validation with MINEMA, blending in CHIRPS and Meteo Rwanda, adding wetness indices, hyper-parameter tuning, and an operational daily dashboard.

Slide 15. Conclusion

To conclude: all five objectives are met on real, open data with a strict temporal hold-out. The framework is accurate, interpretable, agrees with and extends the official flood map, and is fully reproducible. Thank you — I welcome your questions.

Slide 16. Anticipated Panel Questions

Backup answers for likely questions — keep these crisp: circularity is handled by the unobserved noise term; ERA5 is chosen for reproducibility with CHIRPS as future work; the 70% miss is due to a coarse 19% official layer, reframed to enrichment; pressure versus event is a scope decision; and generalisation is shown by testing on a future unseen year.